Harmonic oscillator 2

From the previous section we have for $n = 0, 1, 2, \ldots$

$$\psi_n(x) = C_n \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^n}{dx^n} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

where C_n is the normalization constant

$$C_n = \frac{(-1)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left(\frac{\hbar}{m\omega}\right)^{n/2}$$

Take the derivative of $\psi_n(x)$ to obtain

$$\frac{d}{dx}\psi_n(x) = \frac{m\omega x}{\hbar}\psi_n(x) + C_n \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^{n+1}}{dx^{n+1}} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

Substitute $\psi_{n+1}(x)$ for the right-hand side to obtain

$$\frac{d}{dx}\psi_n(x) = \frac{m\omega x}{\hbar}\psi_n(x) + \frac{C_n}{C_{n+1}}\psi_{n+1}(x)$$

where

$$\frac{C_n}{C_{n+1}} = \frac{\sqrt{2\,(n+1)}}{(-1)} \left(\frac{\hbar}{m\omega}\right)^{-1/2} = -\left(\frac{2m\omega}{\hbar}\right)^{1/2} \sqrt{n+1}$$

Hence

$$\left(\frac{\hbar}{2m\omega}\right)^{1/2} \left[\frac{m\omega x}{\hbar}\psi_n(x) - \frac{d}{dx}\psi_n(x)\right] = \sqrt{n+1}\,\psi_{n+1}(x)$$

By inspection we define step-up operator $\hat{a}^\dagger$ as

$$\hat{a}^\dagger = \left(\frac{\hbar}{2m\omega}\right)^{1/2} \left[\frac{m\omega x}{\hbar} - \frac{d}{dx}\right]$$

Hence

$$\hat{a}^\dagger\psi_n(x) = \sqrt{n+1}\,\psi_{n+1}(x)$$

Let $\hat{a}$ be the step-down operator

$$\hat{a} = \left(\frac{\hbar}{2m\omega}\right)^{1/2} \left[\frac{m\omega x}{\hbar} + \frac{d}{dx}\right]$$

Hence (how to prove?)

$$\hat{a}\psi_n(x) = \sqrt{n}\,\psi_{n-1}(x)$$

Let $\hat{N}$ be the number operator

$$\hat{N} = \hat{a}^\dagger\hat{a}$$

Hence

$$\hat{N}\psi_n = \hat{a}^\dagger\hat{a}\psi_n = \hat{a}^\dagger\left(\sqrt{n}\psi_{n-1}\right) = n\psi_n$$

Eigenmath script